

Web Development

Web Development Fundamentals

Building the Web with CSS, HTML & JavaScript



CSS



HTML



JavaScript

Agenda

- **What We Will Cover**
 - How the Web Works
 - HTML: The Structure
 - CSS: The Style
 - JavaScript: The Interactivity
 - Putting It All Together
 - Tools & Next Steps

How the Web Works

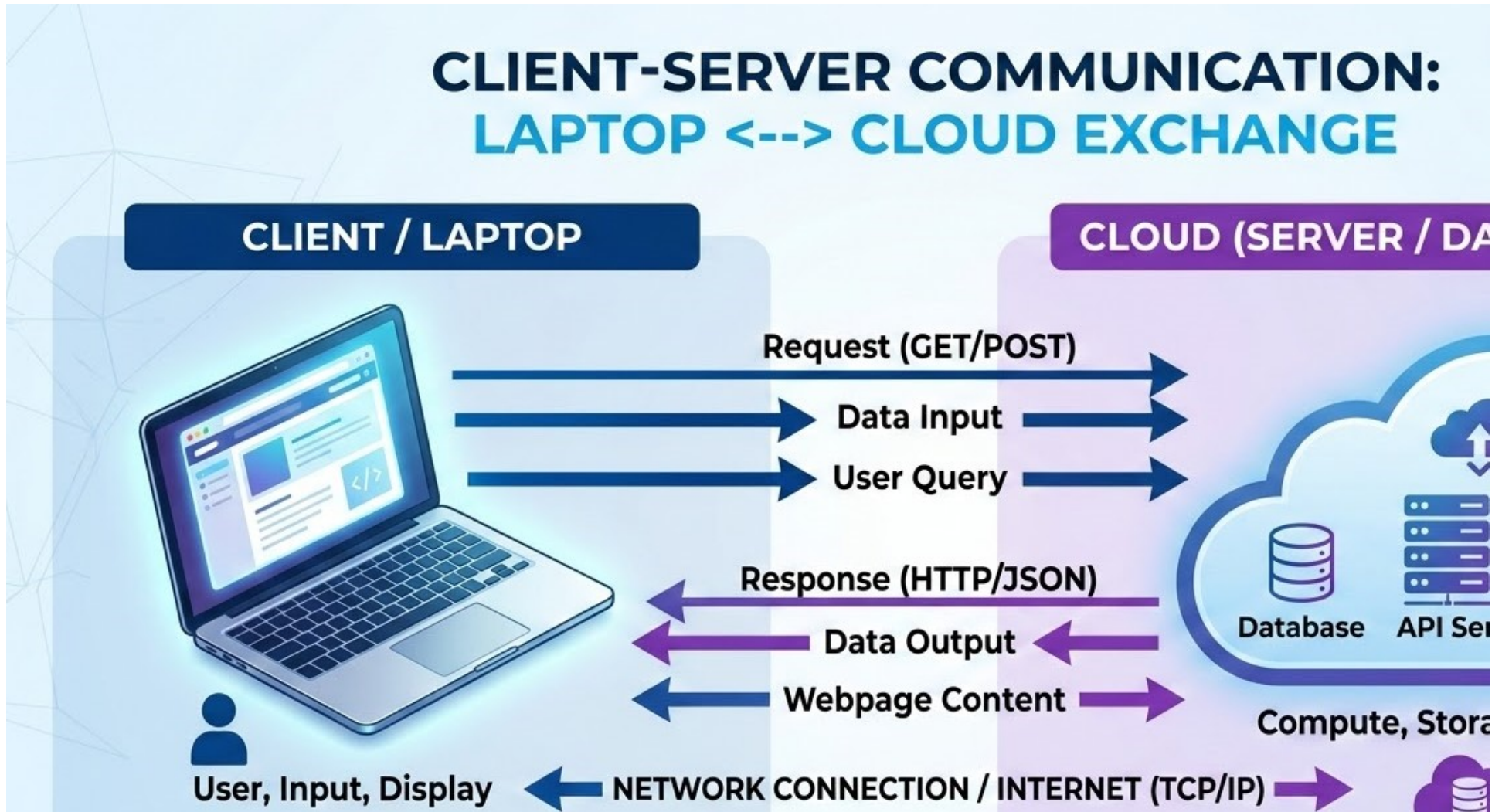
- **Client vs. Server**
 - **Browser (Client):** Requests information (Chrome, Firefox).
 - **Server:** Stores and sends files.
 - **The Code:** HTML, CSS, and JS are sent to your browser to be rendered.

HTML - The Structure

- **The House Analogy**

-  **HTML:** The Framework (Walls, Beams)
-  **CSS:** The Decoration (Paint, Carpet)
- ⚡ **JavaScript:** The Functionality (Electricity, Garage Door)

HTML - The Structure



HTML Basics

- **What is HTML?**

- HyperText Markup Language.
- Uses **Tags** to define elements.
- Tags usually come in pairs: <opening> and </closing>.

```
<p>This is a paragraph.</p>  
<h1>This is a Heading</h1>
```

Common HTML Tags

`<h1>` to `<h6>`: Headings

`<p>`: Paragraphs

`<a href="...">`: Links

``: Images (Self-closing)

`<button>`: Clickable buttons

The HTML Skeleton

- **Standard Document Structure**

```
<!DOCTYPE html>
<html>
  <head>
    <title>My Page</title>
  </head>
  <body>
    <!-- Content goes here -->
  </body>
</html>
```

- `<head>`: Meta info (not visible).
- `<body>`: Visible content.

HTML Demo

- Let's Build Structure

```
<h1>Welcome!</h1>
```

```
<p>Learn to code today.</p>
```

```
<button>Sign Up</button>
```

Intro to CSS

- **Cascading Style Sheets**
 - Controls layout, colors, fonts, and spacing.
 - Separates content (HTML) from design (CSS).
 - Makes sites responsive (mobile-friendly).

CSS Syntax

- **How to Write CSS**

```
selector {  
    property: value;  
}
```

- **Example**

```
h1 {  
    color: blue;  
    font-size: 20px;  
}
```

Selectors

- **Targeting Elements**
 - **Element:** p { } (All paragraphs)
 - **Class:** .btn { } (Reusable, multiple elements)
 - **ID:** #header { } (Unique, one per page)

```
<p class="text">Hello</p>
```

```
<div id="main">Content</div>
```

The Box Model

- **Everything is a Box**
 - **Content:** The text/image.
 - **Padding:** Space inside the border.
 - **Border:** The line around the padding.
 - **Margin:** Space outside the border.

CSS Demo

- Styling the Page

```
button {  
  background: green;  
  color: white;  
  padding: 10px 20px;  
  border: none;  
}
```

JavaScript - The Logic

- **Making it Interactive**
 - A programming language.
 - Runs in the browser.
 - Handles logic, data, and user events.
- **Examples:**
 - Form validation.
 - Image sliders.
 - Fetching data (like tweets or weather).

Variables & Data

- **Storing Information**
 - **Variables:** Containers for data.
 - Keywords: `let`, `var` and `const`
 - **Types:** Strings (text), Numbers, Booleans (true/false).

```
let userName = "Alice";  
  
let age = 25;  
  
let isLoggedIn = true;
```

Functions

- **Reusable Actions**

- A block of code designed to perform a task.
- Can be "called" (run) whenever needed.

```
function sayHello() {  
    alert("Hello World!");  
}
```

The DOM

- **Document Object Model**
 - JS connects to HTML via the DOM.
 - JS can **find** elements.
 - JS can **change** elements.

```
document.querySelector("h1").innerText =  
    "New Title!";
```

Events

- **Listening to the User**
 - JS waits for things to happen.
 - **Click, Scroll, Submit, Load.**

```
button.addEventListener("click", sayHello);
```

Tools of the Trade

- **Where to Code?**
 - **Editor:** VS Code (Recommended), Sublime Text.
 - **Browser:** Chrome or Firefox (Use Developer Tools!).
 - **Practice:** CodePen, JSFiddle (for quick tests).

Next Steps & Resources

- **Keep Building!**
 - **Practice:** Build a personal portfolio.
 - **Docs:** MDN Web Docs (The Bible of Web Dev).
 - **Courses:** freeCodeCamp, The Odin Project.
 - **Q&A**